Redox indices and *Nox1* and *Nox4* mRNA expression in ventrolateral medullary tissue from male rats with atlantoaxial joint disorder

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Summary

We examined regional redox indices and *Nox1*/*Nox4* transcripts in male Wistar rats with atlantoaxial joint disorder. Sham and implant-bearing model rats were sampled on day 28; rats undergoing implant removal on day 28 were sampled on day 56. Ventrolateral medullary blocks containing the rostral ventrolateral medulla and adjacent tissue were analyzed, with six independent rats per group per assay. Model rats showed lower superoxide dismutase activity, higher malondialdehyde and probe-associated fluorescence, and higher estimated *Nox1*/*Nox4* mRNA than sham rats. Restoration-group measurements were generally closer to sham values at the later endpoint. These exploratory findings describe regional biochemical and transcript associations. Different sampling times, absence of a second-surgery sham control, uncertain fluorescence-assay validity in tissue, and incomplete qPCR quality-control records limit interpretation. The results do not establish NOX activity, anatomical specificity, or a sympathetic mechanism.

Key words: Atlantoaxial joint disorder; Ventrolateral medulla; Oxidative stress; NADPH oxidase; RT-qPCR

Introduction

Cervical abnormalities have been associated with hypertension clinically [1], but the underlying relationships remain uncertain. A rat model using a metal implant in the atlantoaxial joint was previously associated with joint instability and elevated blood pressure [2]. Oxidative stress contributes to vascular and autonomic abnormalities in hypertension [3,4,5]. The rostral ventrolateral medulla (RVLM) regulates sympathetic vasomotor tone [6], and experimental studies have linked RVLM redox disturbances and NADPH oxidase (NOX) signaling to neurogenic hypertension [7,8,9,10]. These observations provide a rationale for examining regional redox indices in the atlantoaxial model.

We analyzed superoxide dismutase (SOD), malondialdehyde (MDA), probe-associated fluorescence, and *Nox1*/*Nox4* mRNA in medullary blocks containing the RVLM and adjacent tissue. A restoration group was examined at a later endpoint after implant removal. A related report described Ang II/AT1-R-associated signals in this model [11]; blood-pressure observations were shared with that report, whereas the biochemical and transcript analyses presented here were not reported there. Available H&E images and blood-pressure archives are provided descriptively (Supplementary Figs. S1 and S2). This study characterizes regional associations rather than testing a NOX-dependent sympathetic mechanism.

Methods

Animals and ethics

Healthy clean-grade male Wistar rats, approximately 12 weeks old and weighing approximately 250 g, were supplied by the Laboratory Animal Center of Anhui Medical University (license no. SCXK [Wan] 2017-001). Rats were housed individually in the animal laboratory of the Institute of Acupuncture and Meridian Research, Anhui University of Chinese Medicine, at 22 ± 1°C and 60% relative humidity under an artificial 12 h light/12 h dark cycle, with food and water available ad libitum. Rats acclimatized for 1 week before allocation and surgery. Male rats were selected for consistency with the previous model study [2] and because their body size facilitated the anterior cervical procedure. Applicability to female rats was not evaluated.

All animal procedures were approved by the Animal Ethics Committee of Anhui University of Chinese Medicine (approval no. AHUCM-rats-2022032). The approval covered the 60-rat parent experiment, model induction, foreign-body removal, and terminal sampling. Animal handling followed the Chinese Guidelines on the Humane Treatment of Laboratory Animals.

Study design and sampling

The parent experiment numbered 60 rats and allocated them by random numbers to sham operation (10) or model induction (50). All sham rats survived. During model induction, 6 rats died during surgery and 3 afterward; 41 survived, including 7 with implant loss and 34 with retained implants. After imaging and blood-pressure screening on day 28, successfully modeled rats were assigned by random numbers to model, restoration, and superior cervical ganglion resection groups of 10. The disposition of the remaining 4 rats with retained implants was undocumented. This report includes only sham, model, and restoration rats, initially 10 per group; ganglion-resection and electrophysiological results are not presented.

The implant was removed from restoration rats on day 28. Terminal sampling was on day 28 for sham and model rats and day 56 for restoration rats, without a time-matched second-surgery sham group. Each biochemical endpoint and RT-qPCR included 6 randomly selected independent rats per group, processed separately without cross-animal pooling. Target and reference Cq values were paired within each rat and measured once each. Original animal

identifiers could not be matched to assay samples; numerical record order cannot establish pairing across assays or fully reconstruct premeasurement exclusions. The parent design allowed for preliminary estimates of 18% mortality and 17% implant loss to obtain at least 30 modeled rats. No single primary endpoint or formal outcome-based a priori sample-size calculation was specified.

Model induction and implant removal

The model was induced using the previously described procedure [2]. Rats were anesthetized with 2% inhaled isoflurane at a gas flow of 1 L/min. Anesthetic depth was monitored using respiratory rate and rhythm, the righting reflex, and the response to a noxious stimulus. After shaving and povidone-iodine disinfection, an anterior cervical incision was made, subcutaneous tissue and muscle were separated, and the trachea was displaced to the right to expose the left atlantoaxial joint. With the head rotated to the right, a disinfected S-shaped aluminum implant was inserted into the joint space, secured, and the incision closed. Penicillin sodium was administered intramuscularly at 400,000–800,000 units/kg once daily for 3 days. No postoperative analgesics were administered. Available records did not retain a written rationale or a specific ethics-committee opinion concerning the omission of postoperative analgesia. Food intake, activity, and the wound were checked at least daily for the first 3 postoperative days and 2–3 times per week thereafter. Nine rats died during or after model induction. Reported possible contributing factors were anesthetic overdose, cervical surgical injury, and postoperative infection; individual cause-of-death adjudication records were not retained.

Sham-operated rats underwent the same anterior cervical exposure without implantation. On day 28, implant position and joint morphology were assessed using a Definium-6000 digital X-ray system (GE). Distances between the odontoid process and the left and right lateral masses were measured on images magnified twofold. Model success required a retained implant, asymmetric joint morphology, a left–right distance difference >0.05 mm, and a systolic blood pressure increase of at least 20% relative to baseline with a plateau by week 4.

Restoration rats underwent implant removal on day 28 under the same isoflurane anesthesia. The joint was re-exposed through the original anterior approach, the head rotated rightward, and the implant removed with a needle holder before closure. Postoperative penicillin treatment was unchanged, and no postoperative analgesics were given. Systolic pressure for model screening was measured by tail cuff (IITC Life Science) at baseline and on days 7, 14, 21, and 28. The chamber was prewarmed for 10 min at 30°C; rats rested in a restrainer for 15 min. BPmon Win monitor 1.35 recorded pressure at the onset of the spindle-shaped waveform, and three readings were averaged per rat. These shared source observations [11] were not reanalyzed as a terminal three-group treatment comparison.

Terminal collection and histology

Terminal samples were collected on day 28 in sham and model rats and day 56 in restoration rats. Rats were anesthetized with 2% inhaled isoflurane at 1 L/min for abdominal aortic blood collection, followed by euthanasia with an overdose of inhaled isoflurane. Death was confirmed by the absence of spontaneous breathing, palpable cardiac activity, and withdrawal to a noxious stimulus before brainstem removal. The Paxinos and Watson rat brain atlas and a stereotaxic apparatus guided bilateral tissue collection from the RVLM region, using reference positions 12.12–12.48 mm posterior to bregma, 2.1–2.6 mm lateral, and 10.0–10.4 mm deep. The tissue blocks contained the RVLM and a small amount of adjacent medullary tissue and had a measured mass of approximately 100 mg per rat. They were regional blocks rather than precisely isolated RVLM nuclei, and samples were not pooled across animals.

Histological preparation comprised perfusion with 4% paraformaldehyde in 0.1 mol/l PBS (pH 7.4), postfixation at 4°C for 24–48 h, paraffin embedding, and 4 μm sections, followed by standard hematoxylin–eosin staining. The supplied composite retains tissue and original scale-bar pixels. Specimen counts, image-to-animal correspondence, and field-selection records could not be confirmed; no quantitative histological analysis was performed.

Biochemical assays

Each medullary biochemical assay included 6 independent rats per group. Tissue was minced, ground in liquid nitrogen, lysed on ice in RIPA buffer containing 1 mmol/l PMSF, and centrifuged to collect the supernatant. Total protein concentration was measured using a BCA kit (Beyotime; cat. no. P0010). SOD activity was measured using a WST-1 assay (Nanjing Jiancheng Bioengineering Institute; cat. no. A001-3), with absorbance read at 450 nm. MDA was measured using a thiobarbituric acid (TBA) colorimetric assay (Nanjing Jiancheng Bioengineering Institute; cat. no. A003-1). The retained assay records specified absorbance measurement at 530 nm, whereas the kit manual recommended 532 nm. Both absorbance measurements used an RT-6000 microplate reader (Rayto).

ROS-associated fluorescence was measured using the proprietary green BBoxiProbe® O12 supplied with the BestBio BB-475015 kit, at excitation and emission wavelengths of 488 and 530 nm, respectively. The kit manual specifies fresh serum as the intended sample matrix; the kit was applied to medullary lysates in this study. Records validating its use in the tissue matrix or compatibility with the lysis buffer were unavailable. The fluorescence-reader model was not retained. This measurement was reported as a probe-associated fluorescence index, without assignment to a specific ROS species or absolute ROS concentration.

SOD activity was calculated by dividing the blank-corrected inhibition fraction by 0.5, multiplying by the reaction-to-sample volume ratio (0.24/0.02), and dividing by sample protein concentration (mg/ml); results were expressed as U/mg protein. MDA content was calculated from the blank-corrected sample-to-standard absorbance ratio, multiplied by a standard concentration of 10 nmol/ml, and divided by sample protein concentration; results were expressed as nmol/mg protein. Probe-associated fluorescence was divided by sample protein concentration and reported in protein-normalized arbitrary units (AU). This fluorescence index was not interpreted as an absolute quantity per unit protein mass.

Reverse transcription quantitative PCR

Reverse transcription quantitative PCR (RT-qPCR) analyses included 6 independent rats per group (n = 6). Each target and reference quantification cycle (Cq) was measured once per rat. Target and reference Cq values in the same worksheet column corresponded to the same animal and were paired within that animal. Samples were not pooled. Total RNA was extracted from medullary tissue blocks containing the RVLM using TRIzol reagent (Life Technologies; Thermo Fisher Scientific, Inc.; cat. no. 15596018). Genomic DNA was removed from 1 μg of RNA in a 10 μl reaction at 42°C for 2 min, followed by 1 min on ice. Reverse transcription was performed using PrimeScript RT reagent Kit with gDNA Eraser (Takara Bio, Inc.; cat. no. RR047A) in a 20 μl reaction at 37°C for 15 min followed by 85°C for 5 s. RT-qPCR used NovoStart SYBR qPCR SuperMix Plus (Novoprotein; cat. no. E096-01B) and a PikoReal 96 real-time PCR system (Thermo Fisher Scientific, Inc.). Each 10 μl reaction contained 5 μl of 2× SYBR mix, 1 μl each of forward and reverse primers (10 μmol/l), 1 μl of cDNA, and 2 μl of RNase-free water. Primer sequences were as follows: *Nox1* forward, 5'-GCTCCAGACCTCCATTTG-3'; *Nox1* reverse, 5'-CAGCCCCAACCAAGAAAC-3'; *Nox4* forward, 5'-GGGTTAAACACCTCTGTC-3'; *Nox4* reverse, 5'-GTAGTATTCTGGCCCTTG-3'; β-actin forward, 5'-

CCCATCTATGAGGGTTACGC-3'; and β-actin reverse, 5'-TTTAATGTCACGCACGATTTC-3'. Expected amplicon lengths for *Nox1*, *Nox4*, and β-actin were 84, 95, and 150 bp, respectively. Cycling conditions were 95°C for 1 min followed by 40 cycles of 95°C for 20 s and 60°C for 1 min. Conversion to relative quantities assumed equal amplification efficiencies of approximately 100% for each target and β-actin [12]. Within-animal ΔCq was defined as target Cq minus the paired β-actin Cq. Relative quantity was calculated as $Q = 2^{-\Delta Cq}$. For each gene, this quantity was divided by the arithmetic mean of the relative quantities from the 6 sham-operated rats, giving the sham-operated group an arithmetic mean of 1. Amplification efficiencies and reference-gene stability were not validated. The assay reports mentioned amplification and melting-curve results, but the original curves were unavailable for inspection. Records of no-template controls (NTC) and no-reverse-transcription controls (no-RT) were also unavailable.

Statistical analysis

Statistical analyses used individual rats as experimental units and report mean ± sample SD. All six supplied values per group were included; no outliers were removed or missing values imputed. Repeated measurements did not increase biological n. Biochemical endpoints were compared using Welch's one-way ANOVA and Games–Howell comparisons, with 95% confidence intervals. Shapiro–Wilk tests were diagnostic, with limited sensitivity at n = 6. Adjustment covered the three comparisons within each endpoint, not across the three biochemical endpoints. Tests were two-sided, with $\alpha = 0.05$. Processed inputs, full outputs and software versions, and reproduction code are provided in Supplementary Data 1, File 3, and Appendix 1.

RT-qPCR inference used within-animal ΔCq values and independent-group Welch t tests. All 18 comparisons across the original six-target panel (AT1-R, *Nox1*, *Nox4*, RhoA, ROCK1, and ROCK2) formed one Holm correction family; correction was not restricted to the displayed genes. Supplementary Data 2, File 4, and Appendix 2 provide paired inputs, complete t statistics, degrees of freedom and exact P values, and reproducible code. Animal overlap across assays was not recoverable, so no cross-assay correlations were tested. Archived blood-pressure and H&E materials were displayed without new inferential tests. Model–sham comparisons use day-28 samples; comparisons involving restoration use day-56 samples.

Results

Archived blood pressure and histology

The blood-pressure workbook contained 24 model-labelled, 14 control-labelled, and 10 sham-labelled source rows. The 10 complete sham rows matched the thesis sham summaries, but missing values and repeated identifiers prevented unique reconstruction of the current model cohort. Only four entered restoration summaries remained for weeks 5–8, without animal-level dispersion. These records cannot independently validate the terminal cohorts (Supplementary Fig. S2; Data 3 and 4; File 5). H&E fields show nuclear staining and varied cellular profiles, but uncertain specimen counts and correspondence preclude conclusions about between-group injury or recovery (Supplementary Fig. S1).

Regional biochemical indices

Each medullary biochemical endpoint was analyzed in 6 rats per group. SOD activities in the sham, model, and restoration groups were 1.122 ± 0.072, 0.564 ± 0.072, and 1.035 ± 0.101 U/mg protein, respectively. Corresponding MDA contents were 0.095 ± 0.017, 0.349 ± 0.034, and 0.213 ± 0.049 nmol/mg protein, and ROS-associated fluorescence values were 1326.9 ± 50.6, 3295.6 ± 216.3, and 2265.6 ± 131.6 AU (Fig. 1). Welch's ANOVA indicated

differences among groups: SOD, F(2, 9.816) = 91.626; MDA, F(2, 8.500) = 130.738; and ROS-associated fluorescence, F(2, 7.804) = 314.907 (all P < 0.001). Model–sham mean differences and 95% confidence intervals are presented in Table 1. Compared with model rats sampled on day 28, restoration rats sampled on day 56 had higher SOD activity and lower MDA content and ROS-associated fluorescence (all Games–Howell-adjusted P < 0.001). Compared with sham rats, restoration rats had higher MDA content (adjusted P = 0.0033) and ROS-associated fluorescence (adjusted P < 0.001), whereas the difference in SOD activity was not statistically significant (adjusted P = 0.2534). The absence of a significant SOD difference does not establish equivalence. These restoration-group comparisons describe differences between groups sampled after different durations. Full numerical results are provided in Supplementary File 3.

Nox1 and *Nox4* transcripts

Using paired target and reference measurements within each animal, relative *Nox1* expression in the sham, model, and restoration groups was 1.000 ± 0.134, 1.459 ± 0.156, and 1.180 ± 0.090, respectively. Corresponding *Nox4* expression was 1.000 ± 0.233, 2.902 ± 0.558, and 1.645 ± 0.227 (Fig. 2A and B; n = 6 per group). Both transcripts were higher in model rats than in sham rats (Holm-adjusted P = 0.0037 for *Nox1* and P = 0.0001 for *Nox4*) and lower in restoration rats than in model rats (P = 0.0139 and 0.0037, respectively). Estimated *Nox4* expression was higher in restoration rats than in sham rats (P = 0.0099). The *Nox1* difference between restoration and sham rats did not meet the adjusted significance threshold (P = 0.0618); this does not establish equivalence or restoration of normal expression. Comparisons involving restoration rats span different terminal sampling times. Full results for the family of 18 adjusted tests are provided in Supplementary File 4.

Discussion

The same-time model–sham comparison shows lower SOD activity alongside higher MDA, probe-associated fluorescence, and estimated *Nox1*/*Nox4* transcripts in regional medullary blocks. SOD participates in antioxidant defense, and MDA is a lipid-peroxidation marker [13,14]. The combined observations are consistent with regional redox imbalance, but O12 fluorescence is probe- and matrix-dependent and cannot identify a specific ROS species. The serum-designated kit was applied to RIPA tissue lysates without available validation of matrix or buffer compatibility. Protein-concentration normalization does not convert this index to fluorescence per unit protein mass or correct variation in tissue composition. Differences in sample composition could also influence between-group fluorescence comparisons.

Studies in other hypertension models implicate RVLM oxidative stress in cardiovascular regulation [7,8,10,15]. However, approximately 100 mg regional blocks containing adjacent tissue cannot isolate an RVLM- or cell-specific effect. Higher *Nox1*/*Nox4* mRNA does not demonstrate NOX protein abundance or oxidase activity, and no NOX perturbation or sympathetic nerve recording was performed in this report. Ang II signaling provides a possible context [11,16,17], but Ang II and its receptors were not measured here. An Ang II/AT1-R–NOX–sympathetic sequence therefore remains a hypothesis, not an established pathway in these animals.

Restoration-group values were measured four weeks later than sham and model values. Time, age, postoperative recovery, repeat anesthesia and surgery are confounded with implant removal. Without time-matched and second-surgery sham controls, the group differences cannot establish reversal or an independent implant-removal effect. Surgical implant removal also does not reproduce clinical manual therapy. Postoperative pain or stress may influence

redox and transcript measurements; analgesics were not administered, and the available records lacked a written rationale or a specific ethics opinion addressing that omission. H&E fields with unknown specimen correspondence cannot resolve these uncertainties or demonstrate tissue recovery. The magnitude or persistence of any pain- or stress-related effect at terminal sampling is unknown.

Measurement and source-record limitations further qualify the findings. RT-qPCR used one measurement per target and reference, so technical variability could not be estimated. Amplification efficiency and β-actin stability were unvalidated, and original amplification/melting curves and NTC/no-RT records were unavailable. Relative expression therefore depends on an unverified equal-efficiency assumption. Original animal-to-sample identifiers and instrument outputs were unavailable, limiting verification of measurement quality, sampling and exclusions; processed values and paired Cq inputs support computational reanalysis but do not replace those records. No single primary endpoint or a priori power calculation was specified. Only male rats were studied, and generalizability to females is unknown.

These findings provide exploratory regional biochemical and transcript characterization of the atlantoaxial joint disorder model. Time-matched studies with improved anatomical sampling, assay validation and pathway interventions are needed before attributing the observations to NOX-dependent sympathetic activation or to a therapeutic effect of implant removal.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

Xiangying Yuan: Conceptualization, Investigation, Formal analysis, Writing – original draft. Bin Yang: Methodology, Investigation. Zongbao He: Investigation, Resources. Youkui Lv: Conceptualization, Supervision, Funding acquisition, Writing – review & editing. Zijian Wu: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Data availability

Processed medullary biochemical values, paired RT-qPCR Cq measurements, statistical outputs, and reproducible analysis and plotting scripts are provided in the supplementary material. Archived blood-pressure entries, entered summaries, the unmodified source workbook, and the supplied H&E composite are also provided. Blood-pressure source rows cannot be uniquely mapped to the current terminal cohorts, and H&E specimen counts and image-to-animal correspondence could not be recovered; these materials are presented descriptively. Original instrument outputs, complete assay quality-control records, and original animal-to-sample identifiers were unavailable for independent verification.

DECLARATION on the Use of AI

Codex (OpenAI; accessed October 2026; exact model identifier not archived) assisted with language editing, manuscript organization, methodological review, worksheet checking, statistical reanalysis, and development of conventional analysis and plotting code. Numerical inputs and reproducible scripts are supplied. Experimental data and experimental images were not generated; the H&E tissue and original scale-bar pixels were preserved. The authors reviewed the outputs, figures and conclusions and take responsibility for the work.

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Supplementary material

Supplementary datasets, full statistical outputs, reproduction code, the archived blood-pressure workbook and H&E materials accompany this article. Item descriptions and Supplementary Figs. S1 and S2 are compiled in Yuan_supplementary_information.docx; individual numerical and source files are supplied separately in Supporting_Information.

Tables

Table 1. Differences in ventrolateral medullary oxidative stress markers between model and sham rats sampled at the same time.

Outcome	Model minus sham	95% confidence interval	Adjusted P value
SOD (U/mg protein)	−0.558	−0.672 to −0.443	<0.001
MDA (nmol/mg protein)	0.253	0.209 to 0.298	<0.001
ROS-associated fluorescence (AU)	1968.8	1683.7 to 2253.8	<0.001

Both groups were sampled on day 28, with 6 independent rats per group for each endpoint. Differences retain the original measurement units. Confidence intervals and P values were obtained from the three-group Games–Howell comparisons within each endpoint, without correction across endpoints. Fluorescence is reported in protein-normalized arbitrary units. These results describe group differences in regional tissue blocks and do not establish that NOX expression mediates blood pressure changes.

Figure 1

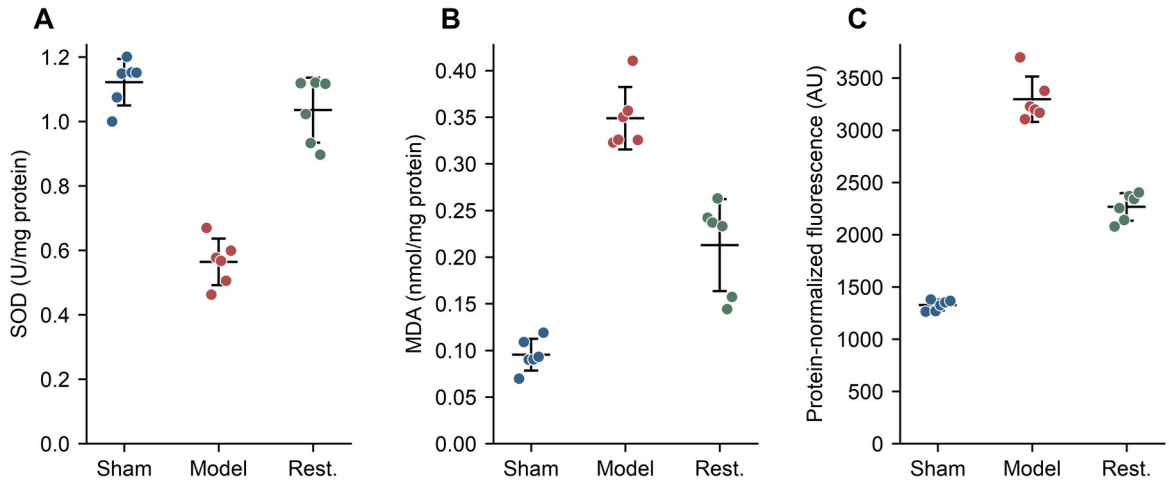


Fig. 1. Oxidative stress markers in ventrolateral medullary tissue blocks containing the RVLM and adjacent tissue. (A) SOD activity; (B) MDA content; and (C) ROS-associated fluorescence. Six independent rats were included per group (n = 6). Each point represents one rat; horizontal lines indicate arithmetic means, and error bars indicate sample SDs. Games–Howell comparisons following Welch's ANOVA are reported in the Results and Supplementary File 3. Multiple-comparison adjustments were applied within each endpoint, but not across endpoints. No significance symbols are shown. Fluorescence is reported as a protein-normalized index in arbitrary units. Sham and model rats were sampled on day 28 and restoration rats on day 56. Sham, sham-operated group; Rest., restoration group; AU, arbitrary fluorescence units; MDA, malondialdehyde; ROS, reactive oxygen species; RVLM, rostral ventrolateral medulla; SD, standard deviation; SOD, superoxide dismutase. Codex (OpenAI) assisted in developing the plotting code.

Figure 2

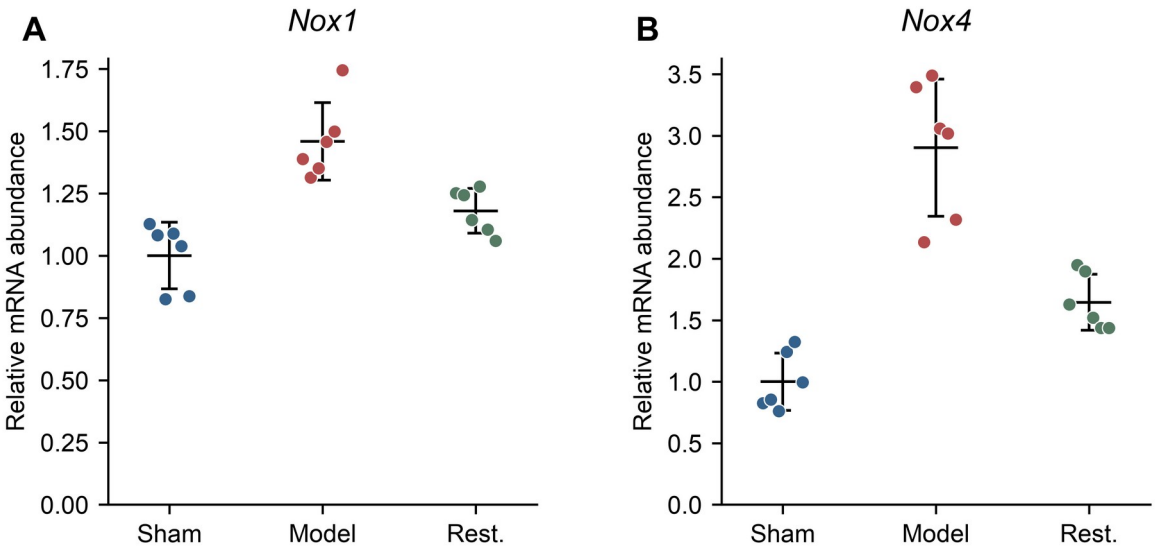


Fig. 2. *Nox1* and *Nox4* mRNA expression in ventrolateral medullary tissue blocks containing the RVLM and adjacent tissue. Relative RT-qPCR expression of (A) *Nox1* and (B) *Nox4*. Six independent rats were included per group (n = 6); tissue samples were not pooled. Each point represents one rat; horizontal lines indicate arithmetic means, and error bars indicate sample SDs. Expression was normalized to β-actin within each rat and then calibrated to the arithmetic mean expression in sham rats. Tests were performed on ΔCq, with a single Holm adjustment applied to all 18 comparisons in the original assay panel. No significance symbols are shown; adjusted P values are provided in the Results and Supplementary File 4. Sham and model rats were sampled on day 28 and restoration rats on day 56. Sham, sham-operated group; Cq, quantification cycle; Nox, NADPH oxidase; RT-qPCR, reverse transcription quantitative PCR; RVLM, rostral ventrolateral medulla; SD, standard deviation. Codex (OpenAI) assisted in developing the plotting code.